JIEMING ZHANG

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EDUCATION

University of Virginia **Charlottesville, VA**

B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA: 3.9*** *Expected May 2029*

Relevant Coursework: Computer Systems Organization I, CPU Architecture, Digital Logic Design, Data Structures & C++, Electric Circuits

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Assembly, Java

Systems & Hardware: CPU Datapaths, Instruction Set Architecture, Digital Logic, Register Files, Stack Operations, Control Sequencing

Development Tools: Vim, Linux, Git, Quartus, ModelSim

Embedded Systems: ESP32, I2C, SPI, UART, CAN Bus, PCB Design

TECHNICAL PROJECTS

Toy ISA Instruction Set & Assembly Programming **Fall 2026**

Computer Systems Organization I | UVA

- Developed and executed programs for a toy 8-bit instruction set, translating previously hand-encoded hexadecimal instructions into assembly source on a university Linux server using Vim.
- Worked with a 4-register architecture, stack and RSP behavior, conditional and unconditional jumps, and register or immediate operand fields selected by instruction code.
- Analyzed instruction encodings comprising a reserved bit, 3-bit instruction code, and operand nibbles for register-register or instruction-specific fields.

CPU Datapath & Controller Design **Spring 2026**

UVA ECE Coursework

- Architected a structural VHDL CPU datapath with ALU functions, register banks, and control sequencers.
- Debugged timing and control-state faults in ModelSim using testbench reset pulses and waveform traces; synthesized the design in Intel Quartus.

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**

Embedded Hardware, Firmware & Python Analytics | Personal Project

- Built non-blocking C++ firmware for an ESP32 to log 100 Hz IMU quaternion and accelerometer data over I2C to SPI MicroSD storage.
- Designed and routed a custom ESP32 PCB with LiPo-to-3.3V power management and IMU/logger interfaces.

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Design schematics and multi-layer PCBs in Altium and KiCad for sensor integration and data logging on an all-electric FSAE racecar.
- Implement synchronized SD card logging from a shared DAQ struct across 17 analog sensor channels.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**

2-person team | Embedded Systems & Wireless Sensing

- Developed C++ firmware on an ESP32 to derive a 0-100 occupancy-density score from Wi-Fi/Bluetooth packet counts and RSSI, calibrated to a nonvolatile empty-room baseline.